IT4010 Research Project – 2025 July Topic Assessment Form

Project ID : 25-26J-400

1. Topic (12 words max)

FitForge AI

2. Research group the project belongs to

AIMS - Autonomous Intelligent Machines and Systems

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Young people today are very busy with school, work, and social lives, leaving them with little time to focus on their health and well-being. This lack of time is causing more young adults to face health problems like obesity, diabetes, heart issues, and mental health challenges such as stress and anxiety. The main reason for these illnesses is that they don't have enough time to take care of themselves properly. With packed schedules, it's hard for them to make time for healthy habits like exercising, sleeping enough, or managing stress, which are important for staying healthy.

Because of their busy lives, many young people skip regular exercise, don't get enough rest, or feel too stressed, which makes them more likely to get sick. Constant pressure to do well in school, succeed at work, or keep up with friends leaves little room for self-care. Also, always being connected to phones or computers takes away time they could use to stay healthy. This leads to weaker bodies, more stress, and a higher chance of serious health problems over time.

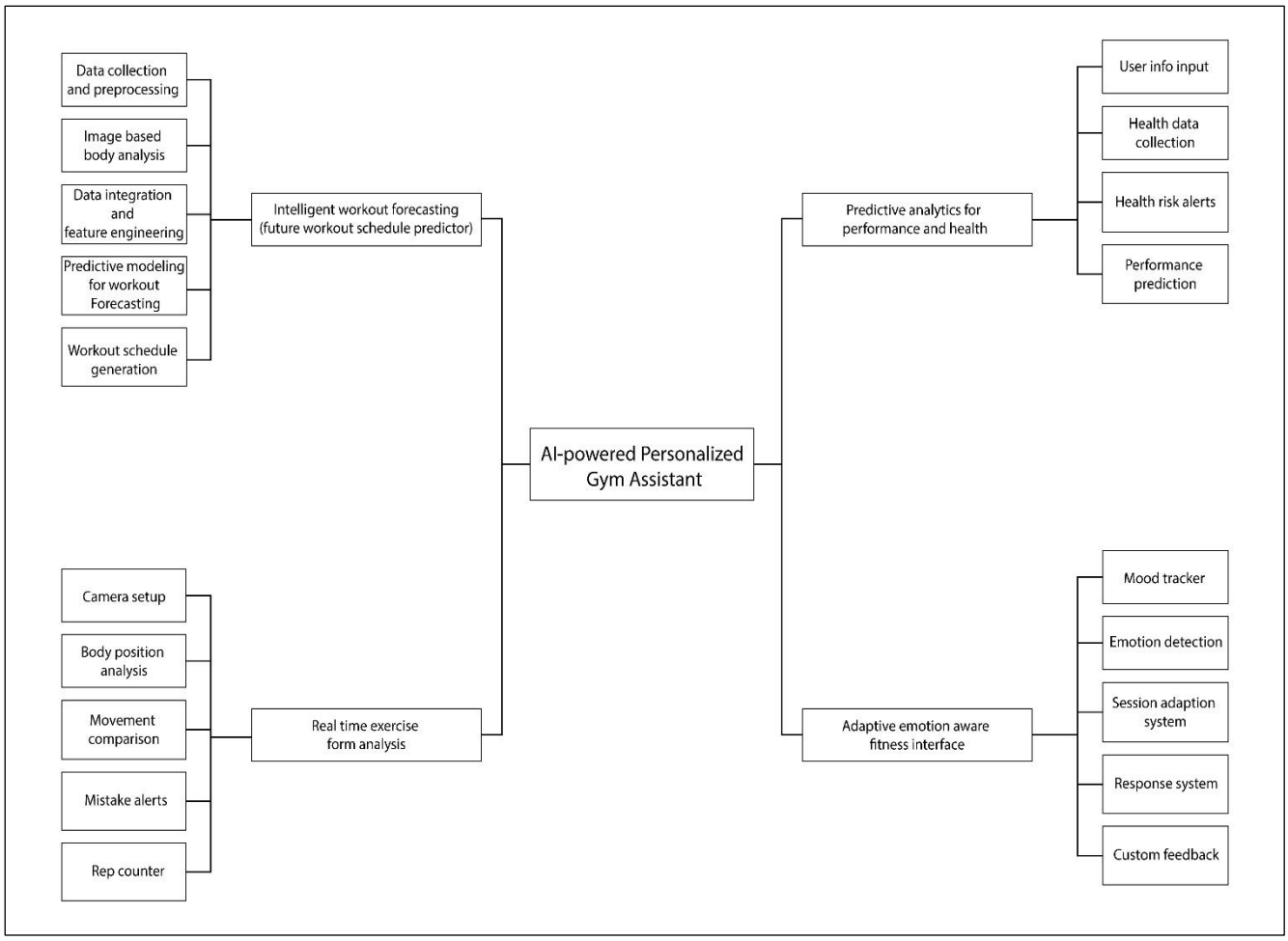
These health issues are a big concern because they can make life harder, increase medical costs, and lower young people's ability to do well in their daily tasks. If this continues, it could lead to even more health problems in the future. This research wants to find out why young people struggle to make time for their health and look for simple ways to help them. The aim is to find easy solutions, like quick exercises or stress-relief tips, that fit

References :
World Health Organization (WHO). (2022). Obesity and overweight – Key facts
Discusses rising obesity rates among youth and contributing lifestyle factors.
Centers for Disease Control and Prevention (CDC). (2022). Physical Activity Facts
Explains the importance of physical activity and trends among adolescents.

Ogden, C. L., et al. (2015). Prevalence of obesity among adults and youth: United States, 2011–2014. *NCHS Data Brief*, (219), 1–8.
PDF link

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

So we find the solution for it. let us tell you about the AI-Powered Personalized Gym Assistant, an awesome solution designed to help busy young adults like us tackle health challenges caused by not having enough time for self-care. With issues like obesity, diabetes, heart problems, and mental health struggles popping up because of our packed schedules, this system offers simple, time-efficient ways to stay healthy.
First up, we've got Predictive Analytics for Performance and Health. This clever feature uses data from wearables—like our heart rate or sleep patterns—to predict health risks and create fitness plans tailored just for us. It helps prevent stuff like diabetes or stress with minimal time investment.
Next, there's the Adaptive Emotion-Aware Fitness Interface. It senses our stress or mood through sensors or inputs and adjusts our workouts accordingly. It might suggest quick stress-relief moves or energizing routines to keep our mental health on track without needing long sessions.
Then, we have Real-Time Exercise Form Analysis. Using our smartphone camera, it checks our exercise form and gives instant feedback to keep our workouts safe and effective, reducing injury risks even during short bursts of exercise.
Lastly, Intelligent Workout Forecasting comes into play. It predicts and schedules short, efficient workouts based on our busy routines, fitting perfectly around school, work, or social plans.
Imagine this: We, as busy young adults, connect with an App/Wearable, which links to an AI Engine handling Analytics, Emotion Detection, Form Analysis, and Forecasting, all tied to a Cloud with our data and schedules. The app grabs real-time info, processes it with AI, and sends us personalized, quick workouts and tips to stay healthy despite our busy lives, lowering the risk of chronic illnesses and stress.


Data collection and preprocessing
Image based body analysis
Data integration and feature engineering
Predictive modeling for workout Forecasting
Workout schedule generation
Intelligent workout forecasting (future workout schedule predictor)
Camera setup
Body position analysis
Movement comparison
Mistake alerts
Rep counter
Real time exercise form analysis
AI-powered Personalized Gym Assistant
Predictive analytics for performance and health
User info input
Health data collection
Health risk alerts
Performance prediction
Adaptive emotion aware fitness interface
Mood tracker
Emotion detection
Session adaption system
Response system
Custom feedback

7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

The AI-Powered Personalized Gym Assistant requires expertise across multiple specialized domains to effectively address the health challenges of busy young adults. Expertise in **sports science** is essential to design fitness plans that optimize performance and prevent injuries, drawing on knowledge of exercise physiology and biomechanics. **Data science** and **machine learning** expertise are critical for developing predictive analytics and intelligent forecasting models, enabling accurate health risk predictions and personalized workout schedules. **Psychology** and **human-computer interaction (HCI)** specialists are needed to create the adaptive emotion-aware interface, ensuring it interprets emotional cues and delivers user-friendly feedback. Additionally, **computer vision** expertise is vital for real-time exercise form analysis, requiring skills in image processing and motion tracking.

Key knowledge areas include understanding health metrics (e.g., heart rate, sleep patterns), emotional recognition techniques, and ergonomic exercise standards. Familiarity with wearable technology and cloud computing is also necessary to integrate and process real-time data effectively.

The solution demands diverse data types to function optimally. **Health data** from wearables (e.g., heart rate, sleep duration, activity levels) is required for predictive analytics. **Emotional data**, collected via user inputs or sensors (e.g., voice tone, facial expressions), supports the emotion-aware interface. **Video data** from smartphone cameras is essential for form analysis, capturing movement and posture. **User profile data** (e.g., age, fitness goals, schedule) and **historical workout data** feed into intelligent forecasting to predict future schedules. All data must be stored securely in a cloud database, ensuring privacy and accessibility. This multidisciplinary expertise and comprehensive data collection enable the assistant to deliver tailored, time-efficient fitness solutions, addressing the research problem of limited self-care time among young adults.

8. Objectives and Novelty

Main Objective : **AI-Powered Personalized Gym Assistant**			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT21374760 – Dissanayake DMKN	Intelligent workout forecasting(future workout schedule predictor)	• Data collection and preprocessing • Image based body analysis • Data integration and feature engineering • Predictive modeling for workout Forecasting • Workout schedule generation	**Image-Based Body Analysis** and its integration with **Predictive Modeling** for scheduling.
IT22595430 – Meegahawala MLHI	Adaptive emotion aware fitness interface	• Mood tracker • Emotion detection • Session adaption system • Response system • Custom feedback	Multimodal emotion and fatigue detection during exercise, dynamically adapting the UI and feedback for personalized experiences.
IT22565662 – Sachinthana GS	Predictive analytics for performance and health	• User info input • Health data collection • Health risk alerts • Performance prediction	Predictive analytics in fitness apps are limited, often focusing on basic trend analysis.
IT21174544 – Mohamed MFI	Real time exercise form analysis	• Camera setup • Body position analysis • Movement comparison • Mistake alerts • Rep counter	Real-time, biomechanically-informed feedback on mobile devices without relying on external sensors or cloud computing.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
IT21374760 – Dissanayake DMKN	This component uses AI to create personalized workout schedules for you based on your details (like age, gender, weight, and fitness goals) and a photo of your body. By analyzing the photo to estimate things like body fat or muscle mass, it predicts the best exercises, sets, reps, and rest days for a set period (e.g., a week or month). The system aims to make your workouts effective, safe, and tailored to your needs, helping you reach your fitness goals faster.
IT22595430 – Meegahawala MLHI	A dynamic, multimodal interface that adapts its visual/audio cues, feedback style, and interaction patterns—based on the user's emotional state, fatigue, and engagement. It enhances motivation, usability, and workout safety while reducing cognitive strain.
IT22565662 – Sachinthana GS	historical workout data, physiological metrics (e.g., heart rate, calories burned), and machine learning to predict future performance, estimate recovery times, and identify potential health issues like overtraining or injury risk. It provides users with data-driven insights to set realistic goals and optimize their fitness journey, as seen in emerging trends in wearable analytics
IT21174544 – Mohamed MFI	Real-Time Exercise Form Analysis is a system that monitors how you perform exercises in real-time to ensure correct technique and prevent injuries. It uses cameras or sensors to track your body movements, analyzes them with AI to detect proper or improper form, and provides instant feedback, like visual or audio cues, to help you adjust your posture or movements immediately.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Mrs.	Anjalie	Gamage	
Co-Supervisor	Mr.	Nelum	Amarasena	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	

b) Does the proposed topic exhibit novelty?

Yes		No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important**:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.